



UNIWERSYTET
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MSCA_DF_10 · Introduction to AI for Financial Applications

Machine Learning in Finance

Reading list

If you have time before the day: skim the paper cards ([Paper_cards.pdf](#)). Optional reading: Mullainathan & Spiess (2017); Gu, Kelly & Xiu (2020), Sections 1–2.1 and Table 1; Krauss, Do & Huck (2017), Sections 3–4. These are marked **optional pre-reading** below.

Course participants received the paper PDFs on the course drive, named by block; those copies are for the course only, so please do not pass them on. The links below go to the published versions.

Opener: prediction vs inference

Mullainathan, S., & Spiess, J. (2017). Machine learning: An applied econometric approach. *Journal of Economic Perspectives*, 31(2), 87–106. <https://doi.org/10.1257/jep.31.2.87> · **optional pre-reading**

Block A: factor models, ML forecasts of returns, and who earns the alpha

Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22. <https://doi.org/10.1016/j.jfineco.2014.10.010>

Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273. <https://doi.org/10.1093/rfs/hhaa009> · **optional pre-reading: Sections 1–2.1 and Table 1**

Avramov, D., Cheng, S., & Metzker, L. (2023). Machine learning vs. economic restrictions: Evidence from stock return predictability. *Management Science*, 69(5), 2587–2619. <https://doi.org/10.1287/mnsc.2022.4449>

Block B: classification, portfolios and frictions

Krauss, C., Do, X. A., & Huck, N. (2017). Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500. *European Journal of Operational Research*, 259(2), 689–702. <https://doi.org/10.1016/j.ejor.2016.10.031> · **optional pre-reading: Sections 3–4**

Block C: credit and the benchmark protocol

Lessmann, S., Baesens, B., Seow, H.-V., & Thomas, L. C. (2015). Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. *European Journal of Operational Research*, 247(1), 124–136. <https://doi.org/10.1016/j.ejor.2015.05.030>

Gunnarsson, B. R., vanden Broucke, S., Baesens, B., Óskarsdóttir, M., & Lemahieu, W. (2021). Deep learning for credit scoring: Do or don't? *European Journal of Operational Research*, 295(1), 292–305.

<https://www.sciencedirect.com/science/article/abs/pii/S037722172100196X>

Fuster, A., Goldsmith-Pinkham, P., Ramadorai, T., & Walther, A. (2022). Predictably unequal? The effects of machine learning on credit markets. *Journal of Finance*, 77(1), 5–47. <https://doi.org/10.1111/jofi.13090>

Exercise only: Zeng, S., Melville, P., Lang, C. A., Boier-Martin, I., & Murphy, C. (2008). Using predictive analysis to improve invoice-to-cash collection. *Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1043–1050.

<https://doi.org/10.1145/1401890.1402014>

Block A+: economics inside the loss

Chen, L., Pelger, M., & Zhu, J. (2024). Deep learning in asset pricing. *Management Science*, 70(2), 714–750.

<https://doi.org/10.1287/mnsc.2023.4695>

Block D: replication and robustness

Harvey, C. R., Liu, Y., & Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68. <https://doi.org/10.1093/rfs/hhv059>

Jensen, T. I., Kelly, B., & Pedersen, L. H. (2023). Is there a replication crisis in finance? *Journal of Finance*, 78(5), 2465–2518. <https://doi.org/10.1111/jofi.13249>

Kelly, B., Malamud, S., & Zhou, K. (2024). The virtue of complexity in return prediction. *Journal of Finance*, 79(1), 459–503. <https://doi.org/10.1111/jofi.13298>

Nagel, S. (2025). Seemingly virtuous complexity in return prediction. NBER Working Paper 34104.

<https://www.nber.org/papers/w34104>

Block E: replication from the author's side

Będowska-Sójka, B., Wójcik, P., & Pele, D. T. (2025). Early warning systems for cryptocurrency markets: Predicting “zombie” assets using machine learning. *North American Journal of Economics and Finance*, 81, 102543. <https://doi.org/10.1016/j.najef.2025.102543>

Extra (block 07, not in the timetable)

Cakici, N., Shahzad, S. J. H., Będowska-Sójka, B., & Zaremba, A. (2024). Machine learning and the cross-section of cryptocurrency returns. *International Review of Financial Analysis*, 94, 103244.

<https://doi.org/10.1016/j.irfa.2024.103244>